

Answer Key — Biology Grade 11

Sub-Strand 1.1: Taxonomy I • Lesson 2: Importance of Classification: Why Do We Need to Identify Organisms Correctly?

Teacher reference only. Do not project or hand out.

Quick marking grid

Q1	Q2	Q3	Q4	Q5	Q6	Q7
D	D	D	D	A	C	B

USE AFTER: PREDICT PHASE — Predict: Think-Pair-Share on 'What could go wrong if people in Kisumu, Nairobi and Kericho identify these organisms differently?'

Q1. A family in Kakamega collects wild mushrooms for supper. They call an edible mushroom and a poisonous look-alike by the same local name. What is the most likely problem?

- A. They will only lose money because the mushrooms cannot be sold
- B. Nothing, because two organisms with the same name must be the same species
- C. Both mushrooms will become safe to eat once they are cooked
- D. They may cook and eat a poisonous mushroom, believing it is safe**

Correct answer: D. If two different mushrooms share one name, people cannot tell which is edible, so someone may eat the poisonous one and get food poisoning. A shared name does not make organisms the same.

USE AFTER: OBSERVE PHASE — Observe: Case A, edible versus poisonous mushrooms (features such as cap, gills, ring, base)

Q2. Learners studied edible and poisonous mushrooms using photographs only. Which approach helps tell an edible mushroom from a deadly look-alike?

- A. Assume all mushrooms growing in the wild are safe
- B. Taste a small piece and see if it makes you sick
- C. Choose the mushroom with the name your family has always used
- D. Compare features such as the cap, gills, ring and base**

Correct answer: D. Identification depends on observing features such as the cap, gills, ring and base. Tasting and assuming all mushrooms are safe are dangerous, and a local name alone may refer to several kinds.

USE AFTER: OBSERVE PHASE — Observe: Case B, useful versus harmful insects (honey bee and ladybird against mosquito, locust and tsetse fly)

Q3. Which statement about insects is correct, based on the honey bee, ladybird, mosquito and locust?

- A. An insect is harmful only if it is large
- B. All insects are pests and should be killed
- C. All insects are useful, so none need to be controlled
- D. Some insects are useful, such as honey bees and ladybirds, while others harm people or crops, such as mosquitoes and locusts**

Correct answer: D. Honey bees pollinate and make honey, and ladybirds help control pests. Mosquitoes spread malaria and locusts destroy crops. Correct identification tells us which insects to protect and which to control.

USE AFTER: OBSERVE PHASE — Observe: Case C, crops versus weeds (maize and beans against couch grass and black jack; herbicide choice)

Q4. A farmer in Kericho must tell maize from couch grass and beans from black jack (*Bidens pilosa*) before weeding or spraying. Why does this identification matter?

- A. Weeds are plants, so they never affect the crop
- B. Black jack and couch grass are useful crops and should be protected
- C. Weeds make the crop taste bitter but do not affect its growth
- D. Weeds compete with crops for light, water and nutrients, and spraying the wrong plant could kill the crop**

Correct answer: D. Weeds compete with crops for light, water and nutrients. A farmer who cannot identify crop and weed correctly may spray or remove the wrong plant and lose the harvest.

USE AFTER: EXPLAIN PHASE — Explain: Discussion and debate 'Is local knowledge enough, or do we also need a universal system?'

Q5. Why are local names (in Kikuyu, Luo, Kiswahili or English) alone not enough for scientists to communicate about organisms across Kenya and the world?

- A. The same organism can have different names in different languages, so people may not be sure they mean the same organism**
- B. Local names are always wrong and local knowledge should be ignored
- C. Local names are too short to describe an organism
- D. Only English names can be used to describe living things

Correct answer: A. Local names are valuable, but one organism may have many names and different organisms may share one name. A common system is needed so everyone is sure which organism is meant.

USE AFTER: EXPLAIN PHASE — Explain: Claim-Evidence-Reasoning (CER) statement 'Why is classification important?'

Q6. A group writes the claim 'Classification is important because it helps us identify organisms safely.' Which statement is the best piece of EVIDENCE to support this claim?

- A. Maize is grown in many parts of Kenya
- B. Classification is very useful to scientists
- C. A poisonous mushroom can look like an edible oyster mushroom, so wrongly identifying it can cause poisoning**
- D. Many Kenyans enjoy eating mushrooms

Correct answer: C. Evidence is a specific fact or case that supports the claim. The poisonous look-alike mushroom shows what can happen when organisms are not identified correctly. The other options are claims or unrelated facts.

USE AFTER: MODEL BUILDING PHASE — Model: revising the class model and adding the 'What is still missing?' box

Q7. Learners agreed that everyone was correct by their own criteria (habitat, colour, usefulness) and that classification helps us identify, study and communicate about organisms. What is still MISSING from the class model?

- A. More local names for each organism
- B. A common system of groups that everyone agrees to use**
- C. More ways to sort organisms by colour and size
- D. More photographs of poisonous mushrooms

Correct answer: B. Different personal criteria and local names cause confusion. The model still needs a shared set of groups that all scientists use, which becomes the next question for the class.